Mahmudul Hasan Khan

Chemical Engineering PhD | Heterogeneous Catalysis | Reaction Engineering | Process Development
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PROFESSIONAL SUMMARY

Heterogeneous catalysis and reaction engineering PhD candidate (April 2027) with 4+ years of hands-on catalyst development, from synthesis and characterization through fixed-bed testing, kinetic modeling, and deactivation diagnosis for renewable-feedstock hydrodeoxygenation (HDO). Developed an ambient-pressure route reaching 85% BTX selectivity at full conversion from a bio-derived feed, diagnosed the root cause of 30–70% activity loss over 16 h on stream, and built a kinetic method that quantifies competitive adsorption in mixed feeds (first author, *ACS Catalysis* 2025). Owns the group's packed-bed reactor, GC/GC-MS, and chemisorption and physisorption instruments; serves as lab safety inspector; mentors undergraduate researchers. Seeking catalysis R&D, reaction engineering, or process development roles.

EDUCATION

Doctor of Philosophy, Chemical Engineering — April 2027 (*Expected*)
Iowa State University (ISU), Ames, IA, USA | GPA: 3.96/4.00

Master of Engineering, Chemical Engineering — May 2025
Iowa State University (ISU), Ames, IA, USA | GPA: 3.96/4.00

Bachelor of Science, Chemical Engineering — February 2021
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh | GPA: 3.62/4.00

TECHNICAL SKILLS

- **Catalysis and Reaction Engineering:** Reaction kinetics and mechanism development; fixed-bed (packed-bed) reactor design, modification, and operation; catalyst deactivation diagnosis and stabilization strategies; transport analysis and kinetic-regime verification (internal and external mass transfer); structure–activity–selectivity relationships.
- **Catalyst Synthesis:** Incipient wetness and wet impregnation; supported metal oxide synthesis and screening; solution-phase reduction of MoO_3; pelletizing and sieving.
- **Characterization (hands-on):** GC (Agilent 6890, 7890, 8890) and GC-MS (Agilent MSD 5575, 5577); chemisorption (Micromeritics AutoChem II: TPR, TPD, pulse titration); BET physisorption (Micromeritics ASAP 2020); PXRD (Rigaku Ultima IV); Raman (Renishaw); FTIR (Bruker 2037); SEM (JEOl JSM-IT500HR); TGA (Netzsch STA449 F1); SEM/TEM sample preparation. **Data analysis:** XPS (CasaXPS); TEM; ICP-OES.
- **Process, Statistics and Data:** Design of Experiments (DoE: full/fractional factorial); Statistical Process Control (SPC); root cause analysis; regression and parameter estimation; Python (NumPy, SciPy, Pandas); MATLAB; JMP; R; MS Excel; Power BI; OriginPro; AI-assisted data automation (Claude).

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — August 2022–Present
Department of Chemical and Biological Engineering (CBE), ISU, Ames, IA, USA

- **Catalyst deactivation and stability:** Owned a time-on-stream deactivation study of MoO_3 during HDO of lignin-derived oxygenates; quantified 30–70% activity loss over 16 h and traced it to over-reduction and coking through spent-catalyst XPS, XRD, TGA, Raman, SEM and TEM.
- **Stabilization strategies:** Designed and tested 6+ redox and carburization pretreatment protocols; showed that none stabilized the catalyst, giving a clear no-go on pretreatment as a mitigation path for this system.
- **Mixed-feed kinetics:** Designed and ran 100+ single- and co-feed fixed-bed experiments (e.g., acetone + anisole; 330 °C, 1 atm H_2) and developed a kinetic framework to extract relative adsorption constants; quantified 6× stronger adsorption of acetone than anisole, evidence that single-feed kinetics can misrepresent performance on real mixed feeds (first author, *ACS Catalysis* 2025).
- **Renewable aromatics process development:** Used DoE to screen 10+ MoO_3-based catalysts for converting 3-methylanisole, a bioethanol fermentation byproduct, to benzene, toluene, and xylenes (BTX), reaching 85% BTX selectivity at full conversion at ambient H_2 pressure; developed a process concept for integrating the route with waste-biomass utilization at existing plants.

- **Catalyst library and selectivity descriptors:** Synthesized and characterized 10+ supported MoO_3 catalysts and benchmarked them for carbonyl HDO; the dataset established support electronegativity and point of zero charge as descriptors for tuning olefin vs. paraffin selectivity (*ACS Catalysis* 2025).
- **Lab ownership and safety:** Primary owner of the group's packed-bed reactor system, GC/GC-MS, and Micromeritics AutoChem II chemisorption and physisorption instruments; modified the reactor system for the co-feed competitive-adsorption study; serve as the group's lab safety inspector.
- **Data automation:** Automated GC peak-area integration with an AI-assisted (Claude) workflow, replacing hours of manual chromatogram processing.
- **Mentoring and collaboration:** Trained and mentored 2 undergraduate researchers and trained fellow graduate students on lab techniques; collaborated with the Roling (computational/DFT) and Huang research groups.
- **Scientific communication:** Lead-authored a $60K USDA-NIFA proposal on catalyst deactivation; leading two first-author manuscripts in preparation (MoO_3 deactivation; 3-methylanisole to BTX); presented at the AIChE Annual Meeting (2024, 2026), North American Catalysis Society Meeting (2025), Great Plains Catalysis Society Symposium (2023-2025); placed 2nd in a departmental research pitch competition.

Teaching Assistant January 2023–December 2023

Department of Chemical and Biological Engineering, ISU, Ames, IA, USA

- Led liquid-liquid extraction and absorption unit-operations labs for 24 senior students (equipment training, lab safety enforcement, report grading) and weekly problem-solving sessions for 30 students in Environmental Chemodynamics.

HSE Intern October 2021–February 2022

Chevron, Dhaka, Bangladesh

- Reviewed internal safety audits and recommended 10+ corrective actions, reducing flagged items in the annual review; updated the regulatory compliance workbook and created monthly “Safety Moment” briefings.

Research Assistant March 2021–September 2021

Institute of Water and Flood Management (IWFM), BUET, Dhaka, Bangladesh

- Analyzed 100+ water and 20+ soil samples to identify pollution trends across two transboundary river basins; co-authored a policy report proposing a national water-quality monitoring framework.

LEADERSHIP AND PROFESSIONAL ACTIVITIES

- **Chemical Engineering Graduate Student Organization, ISU** 2023–2025
  - **Professional Development Chair (2023–24):** Planned, recruited speakers for, and facilitated 8 professional development events, including AI in research, efficient literature review, campus characterization resources, and IP transfer and patenting.
  - **Industry panel:** Organized and facilitated a 5-member industry panel on networking and landing industry jobs.
  - **Vice President (2024–25):** Organized department-wide community events, including a semesterly graduate student picnic, game nights, and socials.
- **Memberships:** American Institute of Chemical Engineers (AIChE); Great Plains Catalysis Society (GPCS); NSF Engineering Research Center for Biorenewable Chemicals (CBiRC); Tau Beta Pi, Iowa Alpha.

SELECTED PUBLICATIONS AND HONORS

- **Khan, M.H.**, Fard, Z.H., Kohler, A.J., Roling, L.T. and Shanks, B.H., 2025. Elucidating the Competitive Hydrodeoxygenation of Lignin-Derived Oxygenates over Bulk MoO_3 Catalyst through Kinetic Analysis. *ACS Catalysis*, 15(17), 15488–15501.
- Kohler, A.J., **Khan, M.H.** and Shanks, B.H., 2025. Identification of Sabatier Descriptors for Hydrodeoxygenation Activity and Selectivity on Supported Molybdenum Oxide Catalysts. *ACS Catalysis*, 15(5), 4317–4330.
- **Khan, M.H.**, Niu, M., Fretag, C., Huang, W., Shanks, B.H., Investigation of the Complex Phenomenon of Molybdenum Oxide (MoO_3) Catalyst Deactivation during Hydrodeoxygenation. 2026 AIChE Annual Meeting.
- 2 additional peer-reviewed reviews (*J. Environ. Manage.* 2023, first author; *RSC Advances* 2025) and 6 conference presentations; full list at mhkhan-saeed.github.io and Google Scholar.
- Second Place, CBE Perfect Pitch Award (3-minute research pitch), ISU May 2026
- George W. Parrott Centennial Graduate Fellowship (selected from 80 applicants), ISU August 2022
- CBE Excellence Fellowship, ISU August 2022